

Structure of Atom

Complete chapter notes — every model and why it failed, all the standard derivations worked in full, the quantum numbers, and the configuration rules with their exceptions.

The spine of this chapter: it is one long argument, not a list of facts. Each model is destroyed by one experiment and replaced by the next — Thomson by **α -scattering**, Rutherford by **classical radiation loss and line spectra**, Bohr by **many-electron atoms and the uncertainty principle**. Learn the chain "model → the one fact it cannot explain → successor" and half the theory questions answer themselves.

1 · Sub-atomic Particles

1.1 Discovery of the electron — cathode rays

In a discharge tube at very low pressure (10^{-4} atm) and high voltage, rays stream from the cathode to the anode.

- They travel in straight lines and cast sharp shadows.
- They are **deflected towards the positive plate** in an electric field $\Rightarrow$ they carry **negative** charge.
- Their properties (charge-to-mass ratio) are **independent of the gas and of the electrode material** $\Rightarrow$ the electron is a universal constituent of all matter.

Thomson (1897) measured the charge-to-mass ratio by balancing electric and magnetic deflections: $\frac{e}{m_e} = 1.758820 \times 10^{11} \text{ C kg}^{-1}$.

Millikan's oil-drop experiment (1909) measured the charge itself, $e = 1.602 \times 10^{-19} \text{ C}$, and showed that charge is always an integral multiple of e — charge is **quantised**. Combining the two gives $m_e = \frac{1.602 \times 10^{-19}}{1.759 \times 10^{11}} = 9.1094 \times 10^{-31} \text{ kg}$.

1.2 Protons and neutrons

Anode (canal) rays, observed with a perforated cathode, are positively charged. Unlike cathode rays, their charge-to-mass ratio **depends on the gas** taken, and it is largest for hydrogen — giving the lightest positive particle, the **proton**. **Chadwick (1932)** bombarded beryllium with α -particles and found an electrically neutral particle of nearly the proton's mass: the **neutron**.

Particle	Symbol	Absolute charge (C)	Relative charge	Mass (kg)	Mass (u)
Electron	e^-	-1.602×10^{-19}	-1	9.1094×10^{-31}	0.00055
Proton	p^+	$+1.602 \times 10^{-19}$	+1	1.67262×10^{-27}	1.00727
Neutron	n	0	0	1.67493×10^{-27}	1.00867

A proton is about **1837 times** heavier than an electron — which is why essentially all of an atom's mass sits in the nucleus.

1.3 Atomic number, mass number, isotopes and isobars

Atomic number Z = number of protons = number of electrons in a neutral atom. Mass number $A = Z +$ number of neutrons. Written as ${}_Z^AX$.

Term	Same	Different	Example
Isotopes	Z (same element)	A (neutron number)	${}^1\text{H}$, ${}^2\text{H}$, ${}^3\text{H}$ · ${}^{35}\text{Cl}$, ${}^{37}\text{Cl}$
Isobars	A	Z (different elements)	${}^{14}\text{C}$ and ${}^{14}\text{N}$
Isotones	Number of neutrons	Z and A	${}^{14}\text{C}$ and ${}^{16}\text{O}$ (8 neutrons each)
Isoelectronic	Number of electrons	Nuclear charge	N^{3-} , O^{2-} , F^- , Ne , Na^+ , Mg^{2+}

FREQUENTLY ASKED

Chemical properties are governed by the **electrons**, so isotopes of an element are chemically almost identical and differ only in physical properties that depend on mass (density, rate of diffusion, rate of reaction in kinetic-isotope situations).

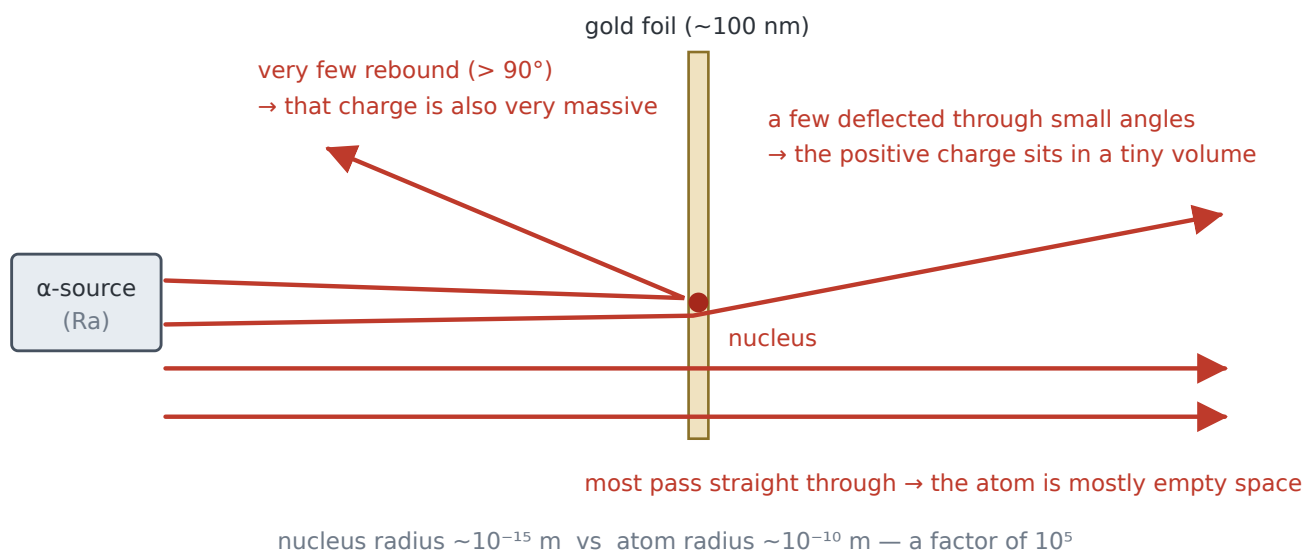
2 · Atomic Models and Their Failures

2.1 Thomson's model (1898)

The atom as a uniform sphere of positive charge with electrons embedded in it — the "plum pudding" or "watermelon" model. It accounted for electrical neutrality and nothing else, and was demolished by α -scattering.

2.2 Rutherford's α -scattering experiment (1911)

A thin gold foil (~100 nm) was bombarded with fast-moving α -particles (He^{2+}) from a radioactive source, detected on a fluorescent ZnS screen.



The α -scattering experiment. Each observation forces one conclusion, and the exam expects them paired: passes through $\rightarrow$ empty space; small deflection $\rightarrow$ positive charge concentrated in a tiny volume; rebound $\rightarrow$ that volume also holds nearly all the mass.

Rutherford's nuclear model. The positive charge and nearly all the mass are concentrated in a minute **nucleus**; electrons revolve around it at high speed in circular orbits; the electrostatic attraction supplies the centripetal force.

THE TWO FATAL DRAWBACKS — LEARN BOTH

1. **Stability.** By classical electrodynamics an *accelerating* charged particle continuously radiates energy. An orbiting electron accelerates, so it should spiral into the nucleus within about 10^{-8} s. Matter would not exist.
2. **Spectra.** A continuously spiralling electron would emit a **continuous** spectrum. Atoms actually give sharp **line** spectra.

It also said nothing about how the electrons are arranged, and so nothing about chemical behaviour.

3 · The Quantum Ideas That Rescued the Atom

3.1 Wave nature of electromagnetic radiation

$c = \nu\lambda$ with $c = 3.0 \times 10^8 \text{ m s}^{-1}$. Wavenumber $\bar{\nu} = \frac{1}{\lambda}$ (cm^{-1}), the number of waves per unit length.

Order of the electromagnetic spectrum by **increasing** frequency (decreasing wavelength): radio < microwave < infrared < **visible** < ultraviolet < X-rays < γ-rays. Visible light runs from about 400 nm (violet) to 750 nm (red).

3.2 Planck's quantum theory (1900)

Classical physics predicted that a black body would radiate ever more strongly at short wavelengths (the "ultraviolet catastrophe"); the observed curve instead peaks and falls. Planck resolved it by abandoning continuous energy:

Energy is emitted or absorbed **only in discrete packets** called quanta, of energy $E = h\nu = \frac{hc}{\lambda}$, with $h = 6.626 \times 10^{-34}$ J s. Emitted energy is always an integral multiple, $n h \nu$.

For light, one quantum is called a **photon**.

3.3 The photoelectric effect (Hertz 1887; Einstein 1905)

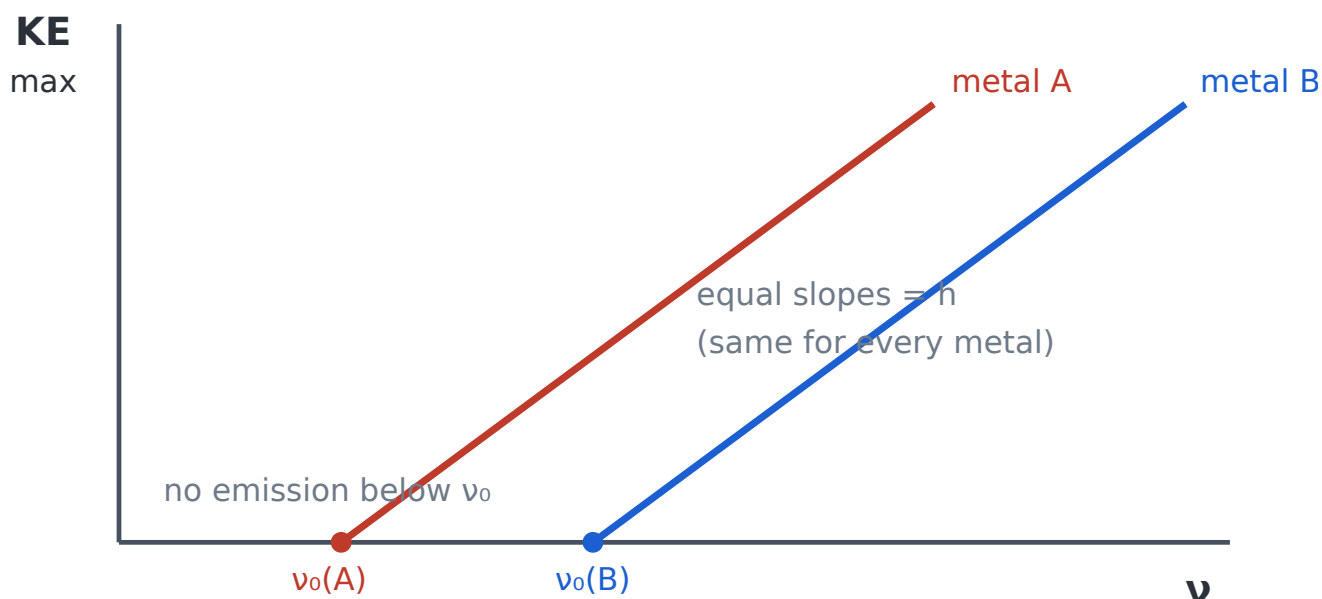
Electrons are ejected from a metal surface when light of sufficiently high frequency falls on it. Four observations, none of which the wave theory can explain:

Observation	Wave theory predicts	Photon theory explains
Below a threshold frequency ν_0 no electron is ejected, however intense the light	Enough intensity should always work	One photon, one electron: if $h\nu < h\nu_0$ the single photon simply cannot pay the price
Emission is instantaneous ($< 10^{-9}$ s)	A time lag while energy accumulates	The transfer is a single collision
Kinetic energy of the electrons rises with frequency , not with intensity	KE should rise with intensity	$E = h\nu$ fixes the energy per photon
Number of electrons (photocurrent) rises with intensity	—	Intensity = number of photons per second

DERIVATION 1 Einstein's photoelectric equation

1. A single photon of energy $h\nu$ is absorbed by a single electron.
2. Part of that energy, the **work function** $\phi = h\nu_0$, is spent freeing the electron from the metal surface.
3. Energy conservation gives the rest to the electron as kinetic energy:
4. $h\nu = h\nu_0 + \frac{1}{2}m_e v^2$.
5. If the electron is stopped by a retarding potential V_0 , then $\frac{1}{2}m_e v^2 = eV_0$, so $eV_0 = h(\nu - \nu_0)$.

$KE_{\max} = h\nu - h\nu_0 = h(\nu - \nu_0)$. **A graph of KE against ν is a straight line of slope h with x -intercept ν_0 — the same slope for every metal, since h is universal.**



Photoelectric graph. Two metals give **parallel** lines: the slope is Planck's constant and cannot change, while the intercept ν_0 measures the work function. Metal B, further right, has the larger work function.

WORKED EXAMPLE 1 — PHOTOELECTRIC NUMERICAL

The work function of caesium is 1.9 eV. Find its threshold frequency and the kinetic energy of an electron ejected by light of wavelength 500 nm.

$$\phi = 1.9 \times 1.602 \times 10^{-19} = 3.04 \times 10^{-19} \text{ J} \Rightarrow \nu_0 = \frac{\phi}{h} = \frac{3.04 \times 10^{-19}}{6.626 \times 10^{-34}} = 4.59 \times 10^{14} \text{ Hz}$$

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{500 \times 10^{-9}} = 3.98 \times 10^{-19} \text{ J}$$

$$KE = 3.98 \times 10^{-19} - 3.04 \times 10^{-19} = 9.4 \times 10^{-20} \text{ J} = 0.59 \text{ eV}$$

3.4 Atomic spectra and the hydrogen spectrum

A **continuous** spectrum (white light through a prism) has no gaps. An **emission line spectrum** is a set of bright lines on a dark background — the fingerprint of the element. An **absorption spectrum** is the exact negative: dark lines at the same positions in a continuous background.

Rydberg's formula for hydrogen: $\bar{\nu} = \frac{1}{\lambda} = 109,677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$, with $n_2 > n_1$.

Series	n_1	n_2	Region
Lyman	1	2, 3, 4 ...	Ultraviolet
Balmer	2	3, 4, 5 ...	Visible
Paschen	3	4, 5, 6 ...	Infrared
Brackett	4	5, 6, 7 ...	Infrared
Pfund	5	6, 7, 8 ...	Far infrared

MEMORY HOOK

Lyman **B**almer **P**aschen **B**rackett **P**fund → "Light **B**ulbs **P**roduce **B**right **P**hotons", with $n_1 = 1, 2, 3, 4, 5$ in that order. Only **Balmer** lies in the visible region.

4 · Bohr's Model of the Hydrogen Atom (1913)

4.1 Postulates

1. The electron revolves around the nucleus in certain **fixed circular orbits** of definite energy, called stationary states or energy shells.
2. While in a stationary state the electron **does not radiate** energy — this is a postulate, introduced precisely to defeat the classical objection.
3. Energy is emitted or absorbed **only when the electron jumps** between stationary states, with $\Delta E = E_2 - E_1 = h\nu$ (Bohr's frequency rule).
4. The angular momentum of the electron is **quantised**: $mvr = n\frac{h}{2\pi}$, $n = 1, 2, 3, \dots$

DERIVATION 2 Radius and energy of the n th Bohr orbit

Part A — radius.

1. The electrostatic attraction between a nucleus of charge $+Ze$ and the electron supplies the centripetal force: $\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{Ze^2}{4\pi\epsilon_0 r}$... (i)
2. Quantisation of angular momentum: $mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr}$... (ii)
3. Substitute (ii) into (i): $m \left(\frac{nh}{2\pi mr} \right)^2 = \frac{Ze^2}{4\pi\epsilon_0 r}$
4. $\frac{n^2 h^2}{4\pi^2 m r^2} = \frac{Ze^2}{4\pi\epsilon_0 r} \Rightarrow r = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}$

$$r_n = 52.9 \frac{n^2}{Z} \text{ pm} = 0.529 \frac{n^2}{Z} \text{ \AA} \text{ — so } r \propto n^2 \text{ and } r \propto 1/Z.$$

Part B — energy.

1. Kinetic energy: from (i), $\text{KE} = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$.
2. Potential energy of the pair: $\text{PE} = -\frac{Ze^2}{4\pi\epsilon_0 r}$ (negative, since the force is attractive).
3. Total energy $E = \text{KE} + \text{PE} = \frac{Ze^2}{8\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r} = -\frac{Ze^2}{8\pi\epsilon_0 r}$.
4. Insert the radius from Part A: $E_n = -\frac{mZ^2 e^4}{8\epsilon_0^2 h^2 n^2}$.

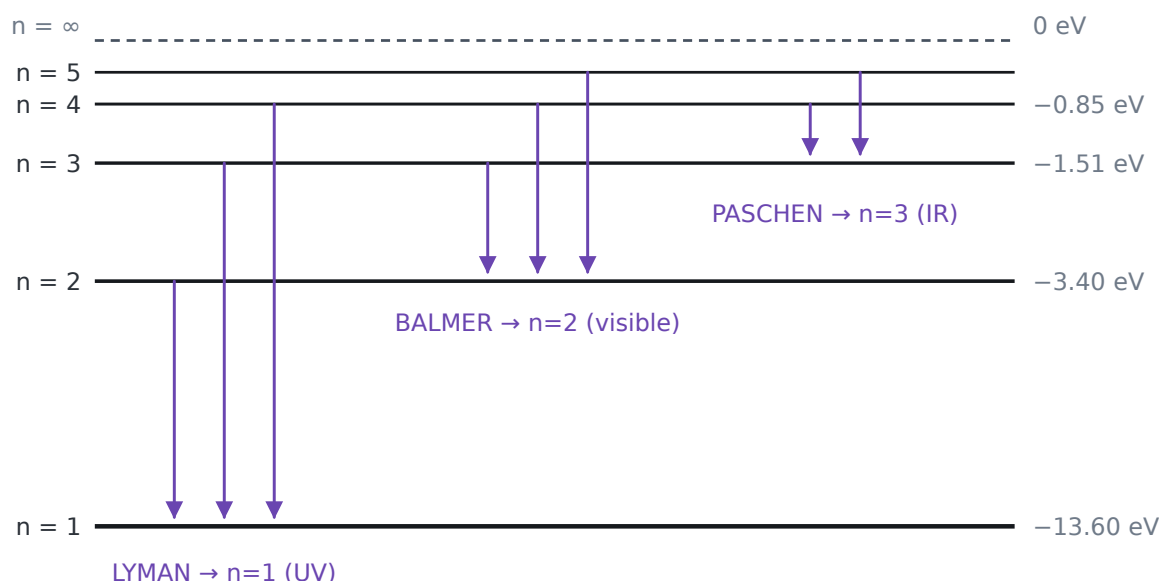
$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J} = -13.6 \frac{Z^2}{n^2} \text{ eV. Note PE} = 2E \text{ and KE} = -E \text{ — so the total energy is negative, which is exactly what "bound" means.}$$

$$\text{Also worth carrying: } v_n = 2.188 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}, \text{ so } v \propto Z/n.$$

DERIVATION 3 The Rydberg formula from Bohr's model

1. An electron falls from level n_2 to level n_1 , emitting one photon: $\Delta E = E_{n_2} - E_{n_1}$.
2. $\Delta E = -2.18 \times 10^{-18} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) = 2.18 \times 10^{-18} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ J.}$
3. By Bohr's frequency rule $\Delta E = h\nu$, so $\nu = \frac{\Delta E}{h} = 3.29 \times 10^{15} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ Hz.}$
4. Wavenumber $\bar{\nu} = \frac{\nu}{c} = \frac{\Delta E}{hc}$.

$$\bar{\nu} = 109,677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1} \text{ — Rydberg's empirical formula, now derived, with } R_H \text{ obtained from } m, e, h \text{ and } \epsilon_0 \text{ alone. This agreement is the reason Bohr's model was accepted.}$$



the levels crowd together as n grows, because $E \propto -1/n^2$ — so every series converges to a limit

Hydrogen energy levels (vertical scale compressed above $n = 3$ so the upper levels stay legible). The spacing between consecutive levels decreases as n rises, so each series has a convergence limit. The series is named by where the electron **lands** (n_1), not where it starts.

WORKED EXAMPLE 2 — THE STANDARD BOHR NUMERICALS

(a) **Radius of the 3rd orbit of Li^{2+} :** $Z = 3$, $n = 3$: $r = 52.9 \times \frac{9}{3} = \mathbf{158.7 \text{ pm}}$.

(b) **Ionisation energy of He^+ :** removing the electron means $n_1 = 1 \rightarrow n_2 = \infty$, so $\text{IE} = 13.6 \times Z^2 = 13.6 \times 4 = \mathbf{54.4 \text{ eV}}$.

(c) **Wavelength of the first Balmer line (H_α , $n_2 = 3 \rightarrow n_1 = 2$):**

$\bar{\nu} = 109677 \left(\frac{1}{4} - \frac{1}{9} \right) = 109677 \times 0.1389 = 15,233 \text{ cm}^{-1} \Rightarrow \lambda = 1/15233 = 6.56 \times 10^{-5} \text{ cm} = \mathbf{656 \text{ nm (red)}}$.

(d) **Number of spectral lines** emitted when an electron drops from the n th level to the ground state = $\frac{n(n-1)}{2}$. From $n = 4$: 6 lines.

LIMITATIONS OF BOHR'S MODEL — A GUARANTEED QUESTION

1. Works **only for one-electron species**: H, He⁺, Li²⁺, Be³⁺. It fails for helium onward because it ignores electron–electron repulsion.
2. Cannot explain the **fine structure** of spectral lines revealed by better spectrometers.
3. Cannot explain the splitting of lines in a magnetic field (**Zeeman effect**) or an electric field (**Stark effect**).
4. Says nothing about the **shapes of molecules or chemical bonding**.
5. It violates the **Heisenberg uncertainty principle** by assigning a definite radius *and* a definite momentum at once — a fixed "orbit" is not allowed.

5 · Towards the Quantum Mechanical Model

5.1 de Broglie's dual nature of matter (1924)

DERIVATION 4 The de Broglie relation

1. For a photon, Planck gives $E = h\nu = \frac{hc}{\lambda}$, and Einstein's mass–energy relation gives $E = mc^2$.
2. Equating: $mc^2 = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{h}{mc} = \frac{h}{p}$.
3. de Broglie's leap: if radiation can behave as a particle, matter can behave as a wave. Replace c by the particle's speed v .

$$\lambda = \frac{h}{mv} = \frac{h}{p}. \text{ For an electron accelerated through } V \text{ volts, } \lambda = \frac{h}{\sqrt{2meV}} \approx \frac{12.27}{\sqrt{V}} \text{ \AA}.$$

Confirmed by **Davisson and Germer (1927)**, who diffracted electrons from a nickel crystal — diffraction is a purely wave phenomenon.

WHY WE NEVER SEE A CRICKET BALL DIFFRACT

λ is inversely proportional to mass. For a 100 g ball at 10 m s^{−1}, $\lambda \approx 6.6 \times 10^{-34}$ m — smaller than any aperture in the universe, so no diffraction is observable. Wave character only matters for particles of very small mass.

Bohr's fourth postulate now stops being arbitrary: a stable orbit must contain a whole number of electron wavelengths, $2\pi r = n\lambda = n\frac{h}{mv}$, which rearranges to exactly $mvr = \frac{nh}{2\pi}$.

5.2 Heisenberg's uncertainty principle (1927)

It is **impossible in principle** to determine simultaneously the exact position and the exact momentum of an electron: $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$, equivalently $\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$.

This is not a limitation of instruments; it follows from the wave nature of matter. Its immediate casualty is the word "orbit": a precise path requires precise position and momentum together.

WORKED EXAMPLE 3 — WHY AN ELECTRON CANNOT RESIDE IN THE NUCLEUS

Confining an electron to a nucleus means $\Delta x \approx 10^{-14}$ m. Then

$$\Delta v \geq \frac{h}{4\pi m \Delta x} = \frac{6.626 \times 10^{-34}}{4\pi(9.11 \times 10^{-31})(10^{-14})} \approx 5.8 \times 10^9 \text{ m s}^{-1}$$

This exceeds the speed of light, which is impossible — so an electron cannot be confined inside the nucleus.

6 · The Quantum Mechanical Model of the Atom

Schrödinger's equation, $\hat{H}\psi = E\psi$, replaces orbits by **wave functions**. Solving it for the hydrogen atom yields the allowed energies E and the wave functions ψ automatically — the quantum numbers fall out of the mathematics rather than being assumed.

ψ itself has no physical meaning; $|\psi|^2$ is the **probability density** — the probability of finding the electron in a unit volume at that point. An **orbital** is the region of space where that probability is high (conventionally the surface enclosing 90% of the probability).

Orbit (Bohr)	Orbital (quantum mechanics)
A definite circular path	A region of probability; no path is defined
Position and momentum known together	Forbidden by the uncertainty principle
Always circular, two-dimensional	Different shapes — spherical, dumbbell, double dumbbell
Holds up to $2n^2$ electrons	Holds at most 2 electrons

6.1 Quantum numbers

Quantum number	Symbol	Allowed values	Tells you
Principal	n	1, 2, 3, ...	Shell; size and (mainly) energy; $2n^2$ electrons at most
Azimuthal (angular momentum)	l	0 to $(n - 1)$	Subshell and shape : $l = 0$ s, 1 p, 2 d, 3 f; orbital angular momentum = $\sqrt{l(l+1)} \frac{h}{2\pi}$
Magnetic	m_l	$-l \dots 0 \dots +l$ ($2l + 1$ values)	Orientation in space; the number of orbitals in the subshell
Spin	m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Direction of electron spin; not a solution of the Schrödinger equation but required by experiment

Orbitals in a subshell = $2l + 1$ (s:1, p:3, d:5, f:7). Electrons in a subshell = $2(2l + 1)$ (s:2, p:6, d:10, f:14). Electrons in a shell = $2n^2$.

6.2 Shapes of orbitals and nodes

s orbitals are spherically symmetric, and the probability never vanishes in any direction. **p orbitals** are dumbbell-shaped, with the two lobes along the x , y or z axis and a **nodal plane** through the nucleus — the probability of finding a p electron *at* the nucleus is exactly zero. Four of the five **d orbitals** (d_{xy} , d_{yz} , d_{zx} , $d_{x^2-y^2}$) are four-lobed clover shapes; d_{z^2} has two lobes along z plus a doughnut in the xy plane.

Nodes are regions of zero probability. Angular (planar) nodes = l · Radial (spherical) nodes = $n - l - 1$ · **Total nodes** = $n - 1$.

WORKED EXAMPLE 4 — COUNTING NODES

For a **3p** orbital: $n = 3$, $l = 1$. Angular nodes = 1; radial nodes = $3 - 1 - 1 = 1$; total = $2 = n - 1$. ✓

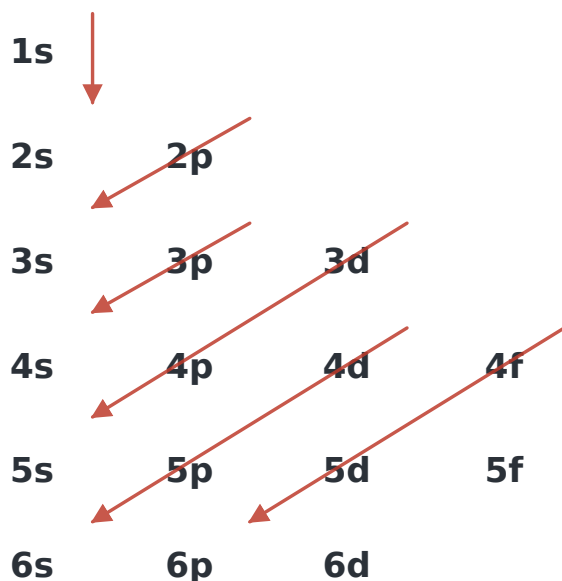
For a **3d** orbital: angular = 2, radial = $3 - 2 - 1 = 0$, total = 2. For a **4s**: angular = 0, radial = 3, total = 3.

6.3 Energies of orbitals

In **hydrogen** (one electron), energy depends on n *only*: $2s = 2p$, and $3s = 3p = 3d$. In **multi-electron atoms**, electron-electron repulsion splits them and the energy depends on **both** n and l .

Two competing effects explain the splitting. **Shielding**: inner electrons screen the nucleus, so an outer electron feels a reduced **effective nuclear charge** Z_{eff} . **Penetration**: within a shell, an s electron spends more time close to the nucleus than p, which does more than d, which does more than f. Greater penetration means less shielding, larger Z_{eff} and lower energy — hence **s < p < d < f** within a shell.

The $(n + l)$ rule. The orbital with the lower $(n + l)$ fills first; if two orbitals have the same $(n + l)$, the one with the **lower** n fills first. Example: 4s ($4 + 0 = 4$) fills before 3d ($3 + 2 = 5$); 3d ($= 5$) fills before 4p ($= 5$) because 3d has the smaller n .



Read along the arrows:

1s 2s 2p 3s 3p 4s 3d 4p

5s 4d 5p 6s 4f 5d 6p 7s

Equivalent to the $(n + l)$ rule:
lower $(n + l)$ fills first; on a tie,
the lower n fills first.

4s fills before 3d — but once
filled, 4s is also emptied first
on ionisation.

The Aufbau (building-up) order. Write the subshells in rows by shell and read along the diagonals. The 4s-before-3d crossover is the single most examined consequence.

6.4 Rules for filling electrons

1. **Aufbau principle.** Electrons occupy the lowest-energy orbital available first, in the $(n + l)$ order above.
2. **Pauli exclusion principle.** No two electrons in an atom can have all four quantum numbers the same — equivalently, an orbital holds at most two electrons and they must have **opposite spins**.
3. **Hund's rule of maximum multiplicity.** Pairing in the orbitals of a given subshell does not begin until each orbital of that subshell has one electron, and all the singly occupied orbitals have **parallel** spins.

6.5 Electronic configurations and the two famous exceptions

Configurations are written by subshell, e.g. O ($Z = 8$): $1s^2 2s^2 2p^4$, or in core notation [He] $2s^2 2p^4$.

Element	Expected	Actual
Chromium ($Z = 24$)	$[\text{Ar}] 3d^4 4s^2$	$[\text{Ar}] 3d^5 4s^1$
Copper ($Z = 29$)	$[\text{Ar}] 3d^9 4s^2$	$[\text{Ar}] 3d^{10} 4s^1$

Reason: exactly half-filled (d^5) and completely filled (d^{10}) subshells have extra stability, because of (i) the **symmetrical distribution** of electron density and (ii) the larger **exchange energy** released when electrons with parallel spins exchange positions — the more such exchanges possible, the more stable the arrangement.

WHERE THIS LOSES MARKS

1. **Electrons are removed from the outermost shell first, which means 4s before 3d** even though 4s was filled first. Fe ($Z = 26$) is $[\text{Ar}]3d^6 4s^2$, but Fe^{2+} is $[\text{Ar}]3d^6$ — *not* $[\text{Ar}]3d^4 4s^2$.
2. Configurations are **written** in shell order (3d before 4s) but **filled** in energy order (4s before 3d). Both conventions are accepted; be consistent.
3. Hund's rule is about a **subshell**, not the whole shell. $2p^3$ is three parallel unpaired electrons, one per orbital.

7 · Where marks are lost

THE TEN TRAPS OF CHAPTER 2

1. **Photoelectric KE depends on frequency; photocurrent depends on intensity.**
No amount of intensity below ν_0 ejects a single electron.
2. **Bohr's model applies only to one-electron species** — H, He^+ , Li^{2+} , Be^{3+} . Using $-13.6Z^2/n^2$ for lithium *atoms* is wrong.
3. **A spectral series is named by the level the electron lands on.** $n_1 = 2$ is always Balmer, whatever n_2 is.
4. **Energy is negative and rises towards zero** as n increases. "Higher energy" means *less negative*, and the gaps get smaller, never larger.
5. **Total nodes** = $n - 1$, of which l are angular. A 3s orbital has 2 radial nodes, not 3.
6. **In hydrogen 2s and 2p are degenerate; in every other atom they are not.**
Questions state which atom deliberately.
7. **m_l counts orientations, not electrons.** A p subshell has $2l + 1 = 3$ orbitals and therefore 6 electrons.
8. **Orbit vs orbital** — an orbital is a probability region and holds at most 2 electrons; only a Bohr orbit holds $2n^2$.
9. **Cr and Cu are the exceptions;** d^4s^2 and d^9s^2 do not occur. Quote symmetry *and* exchange energy as the reason.
10. **Isotopes differ in neutrons, isobars in protons.** "Same mass number, different atomic number" is an isobar — reversing the two is a standard one-mark loss.

Chapter 2 of the rationalised NCERT (2023-24 onward). Constants used: $h = 6.626 \times 10^{-34} \text{ J s}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$, $m_e = 9.11 \times 10^{-31} \text{ kg}$, $e = 1.602 \times 10^{-19} \text{ C}$, $R_H = 109,677 \text{ cm}^{-1}$.